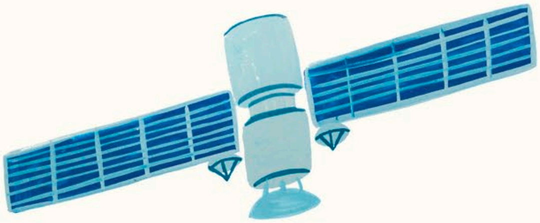
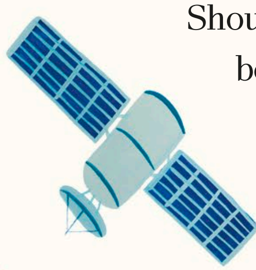
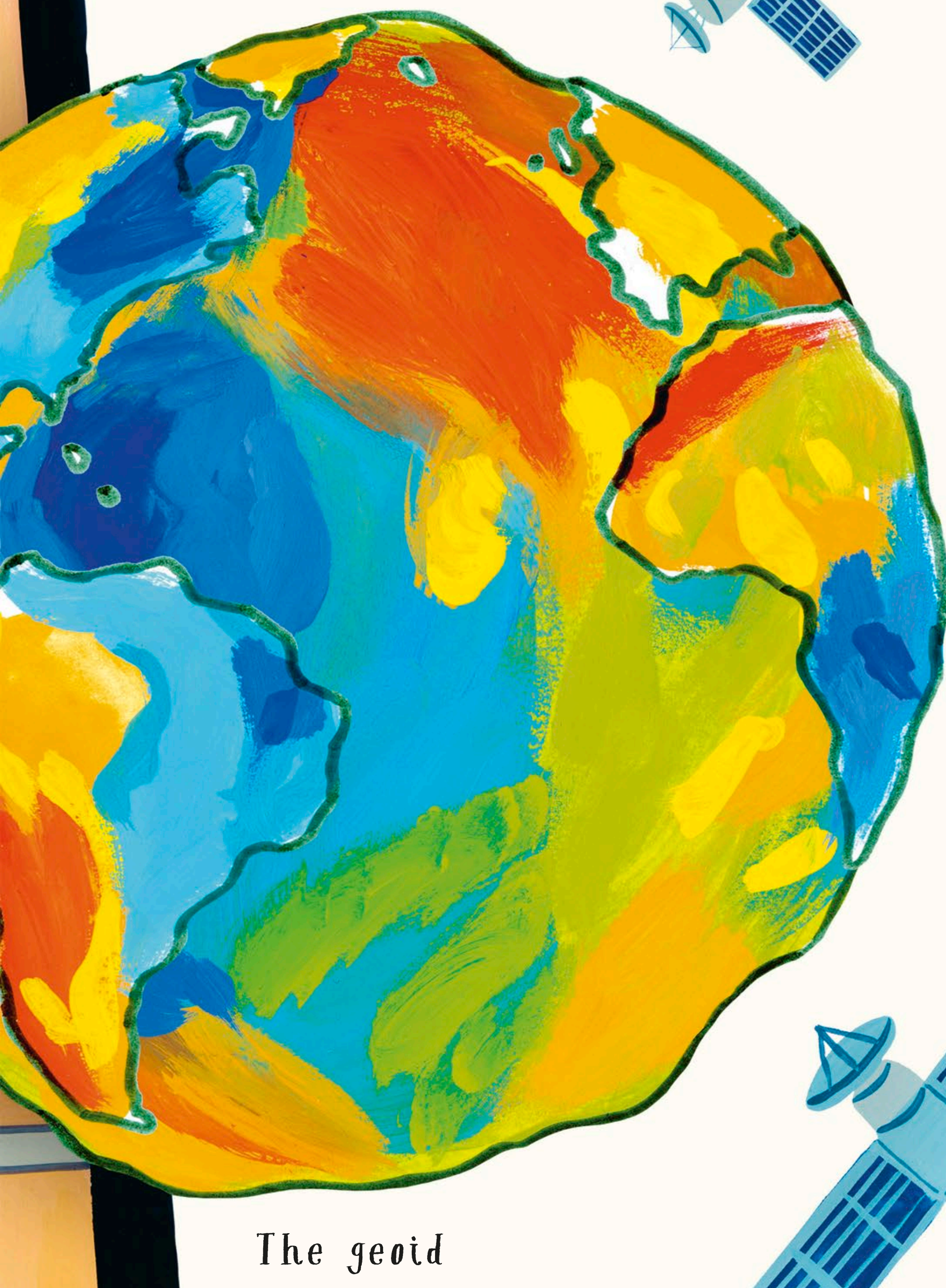


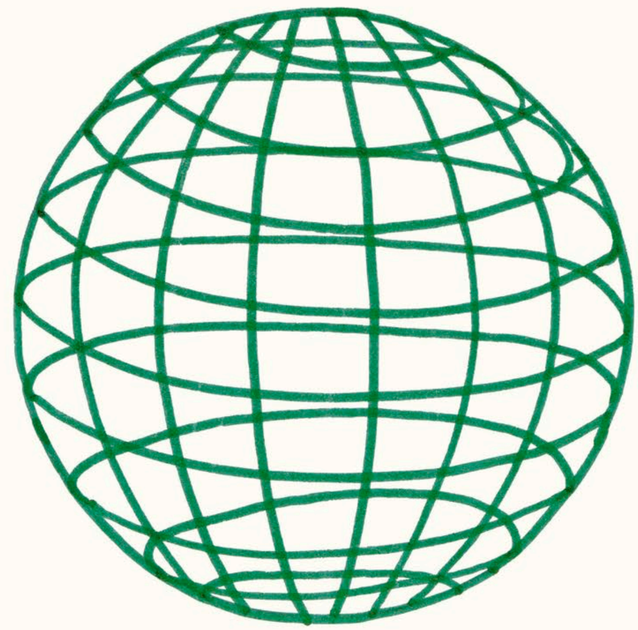
GPS uses a network of 24 satellites in total, each sending out signals at the same time. The time it takes a signal to reach your device tells the GPS how far you are from that satellite.

Uneven planet

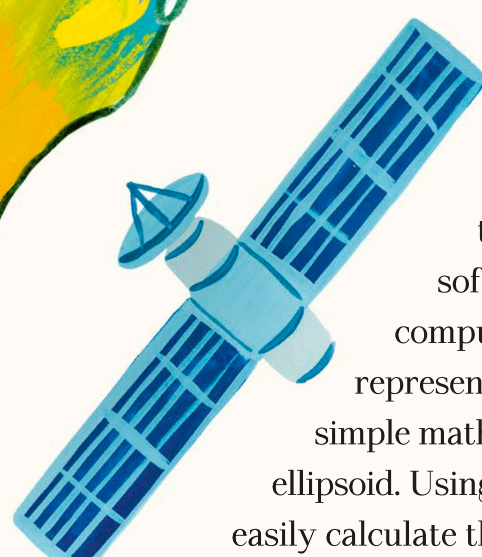
The problem is, Earth has a very bumpy surface and is not a perfect sphere. For example, what counts as zero when we measure the height of a satellite in the sky? Should we use the top of the highest mountain or the bottom of the deepest ocean? What counts as sea level when tides are constantly rising and falling?







An ellipsoid is a mathematical model of the Earth that takes into account changes in altitude (height above sea level), gravity, and tides, and is simple and easy to use.



Gladys helped to solve this problem. She created software that taught the best computers of the time how to represent Earth's unusual shape as a simple mathematical model, called an ellipsoid. Using ellipsoids, computers can easily calculate the position of things on the Earth's surface, which is how GPS works today.

The geoid

For her work, Gladys had to understand the complex shape of the Earth, called the geoid. She used data from satellites to build up an accurate picture of it.